Adding Ridge Regularizer to the model λ=3

Train

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	11
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$\underline{W} = \begin{pmatrix} 0.1 \\ 1.95 \\ 0.03 \end{pmatrix}$$

$$\|W\|^2 = 3.8$$

Test

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	59	78.7	98.4

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	30
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$\underline{W} = \begin{pmatrix} 1.2 \\ 0.72 \\ 0.77 \end{pmatrix}$$

$$\|W\|^2 = 2.45$$

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	34.1	45.1	56.1

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	60
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$\underline{W} = \begin{pmatrix} 1.77 \\ 0.32 \\ 0.99 \end{pmatrix}$$

$$\|W\|^2 = 4.2$$

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	26.2	34.4	42.5